Lab 6 Report

DS5030

Jisung Lee

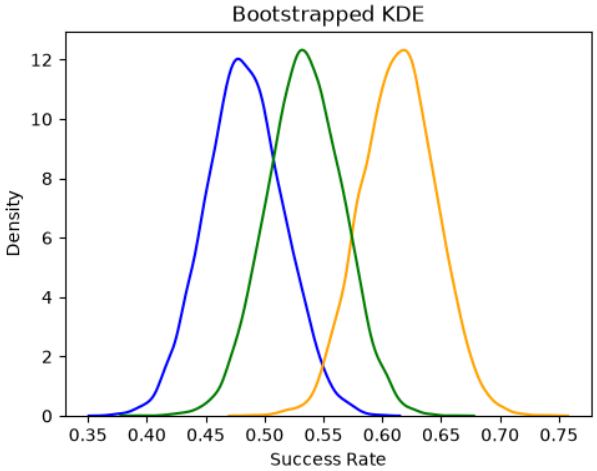
Summary

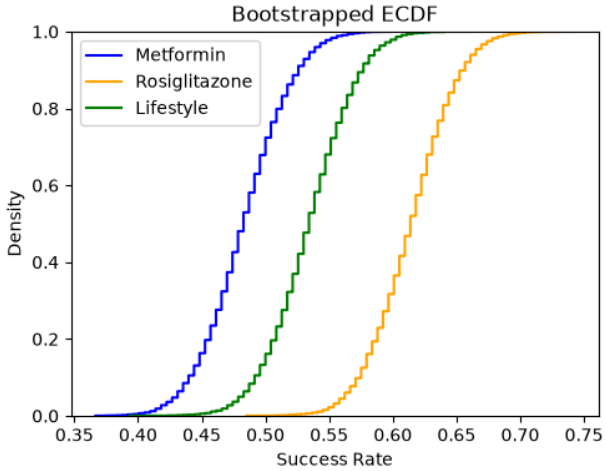
This trial involved 699 pre-diabetic patients and randomized them into groups of metformin, rosiglitazone, and a lifestyle intervention of diet and exercise. Each outcome is recorded as binary, where a success means the patient did not progress to full diabetes over the study's time period. Assignment was random, and the groups are similar-sized, so a difference in group means is an unbiased estimate of each average treatment effect. Values reported come from a bootstrap with 10,000 replications drawn separately from each group.

Outcomes

Treatment	Success	Failure	Success Rate	n
Metformin	112	120	0.483	232
Rosiglitazone	143	90	0.614	233
Lifestyle	125	109	0.534	234

Rosiglitazone had the highest success rate at 61.4%, roughly 13 percentage points over metformin.





Comparisons

Comparison	Point Est.	90% CI	95% CI	99% CI
Met - Rosi	-0.131	(−0.208, −0.058)	(−0.221, −0.041)	(−0.247, −0.015)
Met - Lifestyle	-0.051	(−0.129, 0.026)	(−0.142, 0.039)	(−0.171, 0.069)
Rosi - Lifestyle	0.079	(0.003, 0.157)	(−0.010, 0.169)	(−0.036, 0.195)

The metformin-rosiglitazone comparison is the only statistically significant comparison. The interval excludes zero at the 90, 95, and 99 percent levels, so we reject the null hypothesis of no difference. Rosiglitazone performed significantly better than metformin. There is no evidence that metformin and lifestyle treatments differ, as their intervals contain zero. Rosiglitazone compared to lifestyle is only significant at the 90% level.

Adoption

The evidence is strongest for rosiglitazone. It is clearly superior to metformin and is the best-performing group. It should be preferred over metformin, though the claim that it is preferred over lifestyle intervention requires a larger study. Metformin is the worst-performing group and is significantly worse than rosiglitazone, while almost indistinguishable from lifestyle interventions. There is no evidence to support metformin over either alternative. For the lifestyle intervention, the results are inconclusive against both drugs. Based on this trial evidence, it is neither a better nor a worse option compared to the alternatives.